

Hyper-Casual Cannon Pack – Animated Turrets (URP + Built-in)

Bring life and impact to your mobile or stylized game with this colorful and optimized cannon collection.

This pack includes a set of modular stylized turrets designed for hyper-casual, tower defense, strategy, and action games. Each cannon comes with pre-made shooting animations and is fully optimized for mobile performance.

Perfect for games inspired by Clash-style mechanics, toy battle arenas, and cartoon-style shooters.

Rendering Color Space : Gamma

Formats: FBX, PNG, Unity Prefabs

Materials: Toon_Shader_Outline materials (The custom light shader supports multiple lights only)

shader:

Toon Shader URP and BuiltIn

User Manual

1. Introduction

This shader provides a professional Toon / Cartoon look inside Unity URP OR BuiltIn

It can be used for:

- Stylized characters
- Cartoon environments
- Hyper casual games
- Mobile and PC projects

Features include:

- Toon lighting
- Outline rendering

- **Fog support**
- **Ambient occlusion**
- **Rim lighting**
- **MatCap shading**
- **Shadows**
- **Mobile optimization**

**No scripts are required.
Only a Material is needed.**

2. Setup

Creating a Material

1. **Right click in the Project window**
 2. **Create → Material**
 3. **Select the shader: Toon_BuiltIn_Outline
OR
Toon_URP_Outline**
 4. **Assign the material to a mesh.**
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3. Basic Usage (Quick Start)

To get a fast toon look:

- **Assign a Base Map texture**
- **Set Highlight Color**
- **Set Shadow Color**
- **Ramp Threshold ≈ 0.7**
- **Ramp Smoothing ≈ 0.1**

This creates a clean toon shading.

4. Lighting Controls

Ramp Threshold

Controls how much of the surface is lit vs shadowed.

- **Lower value = more light**
- **Higher value = more shadow**

Ramp Smoothing

Controls the softness of the transition.

- **0 = sharp cartoon edge**
- **Higher = smoother gradient**

5. Outline

Enabling Outline

- **Enable Use Outline**
- **Choose Outline Color**
- **Adjust Outline Width**

Tips

If the outline is not visible:

- **Increase Outline Width**
- **Try a value like 0.02 for testing**

Recommended ranges:

- **Small props: 0.003 – 0.01**
- **Characters: 0.006 – 0.02**
- **Large objects: 0.02+**

6. Ambient Occlusion

SSAO (High Quality)

Adds soft contact shadows between objects.

To enable:

- 1. Open URP Renderer Data**
- 2. Add Renderer Feature**
- 3. Choose Screen Space Ambient Occlusion**
- 4. Enable the feature**
- 5. In the material:**

Screen Space AO = ON

Mobile Fake AO

A lightweight AO alternative for mobile.

- Enable Mobile Fake AO**
- Adjust AO Intensity as needed**

Provides depth at minimal performance cost.

7. Rim Lighting

Adds glow on object edges.

- Enable Use Rim**
- Adjust Rim Min / Max**

Ideal for stylized characters.

8. MatCap

Adds stylized baked lighting.

- **Enable Use MatCap**
- **Assign a MatCap texture**

Great for anime/cartoon styles.

Not recommended for mobile performance.

9. Emission

Makes objects glow.

- **Enable Use Emission**
 - **Assign Emission Map**
 - **Choose Emission Color**
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10. Fog

The shader automatically supports URP fog.

If fog is not visible:

- **Ensure fog is enabled in URP**
 - **Confirm the camera uses the correct URP renderer**
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11. Mobile Optimization

For best mobile performance:

- **Mobile Mode = ON**

- **SSAO = OFF**
 - **Mobile Fake AO = ON**
 - **MatCap = OFF**
 - **Metallic Map = OFF**
 - **Emission only if needed**
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12. Common Problems

Outline not visible

- **Use Outline must be ON**
 - **Increase Outline Width**
 - **Check object scale**
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SSAO not working

- **Renderer Feature not enabled**
 - **Screen Space AO is OFF**
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Low performance

- **Enable Mobile Mode**
 - **Disable MatCap**
 - **Disable SSAO**
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13. Recommended Presets

Stylized Character

- **Outline ON**
- **Rim ON**
- **MatCap ON**

- **SSAO ON**
- **Low ramp smoothing**

Hyper Casual Mobile

- **Outline ON**
- **Mobile Mode ON**
- **Fake AO ON**
- **SSAO OFF**
- **MatCap OFF**

Anime Style

- **Outline ON**
- **Ramp Smoothing = 0**
- **Rim ON**
- **MatCap ON**